

Robustness of Hierarchy-Aware CLIP Adaptation Under Noisy Taxonomic Supervision

Submission Report

GitHub repository: <https://github.com/manuvikash/VLM-finetune>

Abstract. Vision-language models are often adapted to domain-specific recognition tasks using labels that are organized in a taxonomy. In practice, these taxonomies are rarely perfect: parent labels can be missing, ambiguous, inconsistently mapped across sources, or systematically wrong. This paper studies how hierarchy-aware adaptation of CLIP behaves when parent-class supervision is corrupted. We construct a three-level CIFAR-100 hierarchy with 100 leaf classes, 20 parent classes, and six top-level domains, then compare zero-shot CLIP, flat supervised adaptation, clean hierarchy-aware adaptation, and noisy hierarchy-aware adaptation. To make a multi-seed degradation study feasible on a commodity 4GB GPU, we cache OpenCLIP ViT-B/32 image embeddings and train a lightweight residual adapter with leaf and parent classification heads. The hierarchy objective combines direct parent classification with a differentiable leaf-to-parent consistency loss, allowing parent supervision to influence the fine-grained decision boundary. Across three seeds, flat adaptation reaches 73.15% leaf accuracy and clean hierarchy-aware adaptation reaches 73.20%. Uniform parent-label corruption is relatively benign, retaining 71.19% leaf accuracy even at full corruption. However, structured taxonomy noise is far more damaging: in-domain corruption drops to 63.97%, while cyclic systematic corruption drops to 47.93%. These results show that the danger in hierarchical supervision is not only the amount of taxonomy noise, but whether the noise is coherent enough to teach the model a wrong ontology.

Keywords: vision-language models · CLIP · hierarchical classification · noisy labels · taxonomy · robustness · reproducibility

1 Introduction

Large vision-language models (VLMs) such as CLIP are widely used as foundation models for downstream visual recognition. A common adaptation strategy is to freeze or lightly tune the visual backbone, then train task-specific heads on top of the learned image representation. Many real classification tasks are not flat, however. Product catalogs, species datasets, medical image ontologies, and remote-sensing taxonomies all organize labels into a hierarchy. For example, an image may be labeled as a fine-grained leaf class such as “pickup truck,” a parent class such as “vehicle,” and a broader top-level domain such as “object” or “scene.”

Hierarchical supervision can provide useful inductive bias. If a model confuses two leaf classes under the same parent, the error is usually less severe than confusing classes across unrelated parents. A parent-level loss can encourage features that respect this structure and may improve calibration, top-k behavior, or robustness when leaf labels are sparse. Yet this benefit comes with a practical risk. Real-world taxonomies are often noisy. Parent classes may be assigned by separate annotators, inherited from legacy data sources, or merged from incompatible schemas. In such cases, hierarchy-aware training can inject supervision that conflicts with the leaf task.

This paper studies the robustness of hierarchy-aware CLIP adaptation under controlled taxonomy noise. The goal is not simply to show whether hierarchy helps on clean CIFAR-100, but to identify when hierarchical supervision becomes harmful. We focus on three questions:

1. Does hierarchy-aware adaptation improve over flat supervised adaptation?
2. How does leaf accuracy degrade as parent-label corruption increases?
3. Are structured taxonomy errors more damaging than random parent noise?

The main contribution is an end-to-end reproducible experiment pipeline that builds a CIFAR-100 hierarchy, runs multi-seed noise sweeps, exports paper-ready figures and CSV tables, and provides a Springer-format final report. The empirical finding is clear: hierarchy-aware adaptation is robust to diffuse uniform parent noise, but vulnerable to coherent systematic noise. The expanded analysis also shows that calibration, semantic error distance, parent-level collapse, and per-parent sensitivity reveal failure modes that are not visible from leaf accuracy alone.

2 Related Work

CLIP introduced contrastive language-image pretraining at web scale, enabling strong zero-shot transfer through natural language prompts [1]. OpenCLIP provides open implementations and pretrained checkpoints that make CLIP-style models practical for research use [2]. Because full fine-tuning of VLMs can be expensive, downstream adaptation often uses linear probing, adapters, prompt tuning, or partial fine-tuning. These methods trade off computational cost and task-specific flexibility.

Hierarchical classification has a long history in computer vision and machine learning, with surveys emphasizing that the hierarchy can affect both model training and evaluation [5]. Instead of treating all errors equally, hierarchical methods exploit relationships among labels. Some approaches modify the loss, some constrain predictions to respect a tree, and others evaluate predictions with hierarchy-aware distances. A key motivation is that semantic structure can regularize learning and provide more informative error signals than flat classification alone.

Noisy-label learning is also closely related. In standard noisy-label settings, the class label itself may be corrupted, and prior work has proposed loss correction methods that estimate or compensate for label-transition noise [6]. This

work differs in that the leaf label remains clean while the auxiliary parent label is corrupted. The setting is common in practice: a fine label may be curated manually while higher-level metadata is inherited from a weak or inconsistent taxonomy. This makes the problem especially subtle, because the primary target is correct but the auxiliary supervision can still damage the learned representation. The structured noise conditions also connect to shortcut-learning analyses: a model can exploit a spurious but coherent signal when it is predictive during training, even if the signal encodes the wrong abstraction [7].

3 Methodology

3.1 Dataset Construction

The data pipeline converts CIFAR-100 into a three-level hierarchy. CIFAR-100 contains 100 fine classes and 20 coarse superclasses [3]. We use each fine class as a leaf label and each coarse superclass as a parent label. A top-level domain is added manually for each parent group, yielding paths of the form:

$$\text{domain} \rightarrow \text{parent} \rightarrow \text{leaf}. \quad (1)$$

The full official CIFAR-100 split is used: 50,000 training images and 10,000 test images. This avoids ambiguity introduced by random train-test splits and makes the experiment easier to reproduce.

3.2 Model Architecture

The visual backbone is OpenCLIP ViT-B/32 with OpenAI pretrained weights. In the main sweep, image embeddings are extracted once and L2-normalized. Cached features allow many taxonomy-noise configurations to be evaluated without repeatedly running the vision transformer. This design is computationally important for commodity hardware, but it still allows a trainable shared representation through a residual adapter:

$$h(x) = z(x) + \text{LN}(W_2 \text{GELU}(W_1 z(x))), \quad (2)$$

where $z(x)$ is the frozen CLIP image embedding. The adapter output $h(x)$ is fed to a leaf head. In hierarchy-aware runs, it is also fed to a parent head. The flat model and hierarchy-aware model therefore share the same feature cache and adapter family; the difference is the presence or absence of the hierarchy loss.

3.3 Training Objectives

The flat adaptation objective is standard leaf cross entropy:

$$\mathcal{L}_{flat} = \text{CE}(y, \hat{y}), \quad (3)$$

where y is the leaf label and $\hat{y}$ is the leaf prediction.

The hierarchy-aware objective adds parent supervision:

$$\mathcal{L} = \text{CE}(y, \hat{y}) + \lambda [(1 - \alpha)\text{CE}(p, \hat{p}) + \alpha\text{CE}(p, g(\hat{y}))]. \quad (4)$$

Here p is the parent target, $\hat{p}$ is the parent-head prediction, and $g(\hat{y})$ aggregates leaf logits into parent logits using parent-wise log-sum-exp. This aggregation makes the auxiliary parent target act directly on the leaf decision boundary. Without this coupling, a frozen-feature model with independent heads can absorb parent-label noise in the parent head while leaving leaf predictions almost unchanged. We use $\lambda = 1.0$ and $\alpha = 0.8$ in the final sweep.

3.4 Taxonomy Noise Models

Only training parent labels are corrupted. Leaf labels, validation targets, and test targets remain clean. Three noise mechanisms are studied:

- **Uniform noise:** with probability ρ , replace the true parent with a uniformly sampled different parent.
- **In-domain noise:** with probability ρ , replace the true parent with another parent that shares the same top-level domain when such a parent exists. This simulates plausible taxonomy confusion among related groups.
- **Cyclic noise:** with probability ρ , replace the true parent with the next parent in a deterministic sorted cycle. This simulates a coherent schema mapping error.

The corruption fraction ρ is swept over $\{0.0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$.

4 Experiments

4.1 Baselines

Four primary conditions are evaluated. The first is zero-shot CLIP, using text prompts of the form “a photo of a {class}.” The second is flat supervised adaptation, which trains the residual adapter and leaf head with leaf labels only. The third is hierarchy-aware adaptation with clean parent labels. The fourth is hierarchy-aware adaptation with corrupted parent labels.

4.2 Hyperparameters

All supervised adapter experiments are run with three seeds: 42, 123, and 456. The adapter is trained for 30 epochs with AdamW, learning rate 10^{-3} , weight decay 10^{-4} , batch size 8192, adapter width 256, and dropout 0.05. The feature extraction batch size is 256. All experiments are run with Weights & Biases disabled and write deterministic JSON rows to `runs/results.json`. In addition to aggregate metrics, the rerun used for this report logs `runs/per_parent_metrics.csv`, which records leaf and parent accuracy for each true parent group, seed, corruption type, and noise fraction.

4.3 Metrics

We report six metrics. Leaf accuracy is the primary fine-grained top-1 metric. Top-5 accuracy measures whether the correct leaf appears among the five most likely predictions. Parent accuracy measures whether the predicted parent is correct, either from the parent head or from the predicted leaf’s canonical parent. Hierarchical distance measures tree distance between predicted and true leaves: 0 for correct leaf, 2 for same-parent mistakes, 4 for same-domain mistakes, and 6 for cross-domain mistakes. Severity-weighted accuracy gives 1.0 credit for a correct leaf, 0.5 credit for a wrong leaf under the same parent, and 0.0 otherwise. Expected calibration error (ECE) is computed with 15 equal-width confidence bins [4].

5 Results

5.1 Main Results

Table 1 summarizes the main comparison. Zero-shot CLIP obtains 56.05% leaf accuracy. Both supervised adapters provide a large gain, reaching approximately 73% leaf accuracy. Clean hierarchy-aware adaptation is statistically similar to flat adaptation in leaf accuracy, but it improves hierarchical distance, severity-weighted accuracy, and calibration. Under 20% uniform parent noise, leaf accuracy falls modestly to 72.56%, but ECE increases from 0.0179 to 0.0692. This indicates that mild taxonomy noise may leave top-1 predictions mostly intact while still degrading confidence quality.

Table 1. Main results on the full CIFAR-100 hierarchy. Mean $\pm$ standard deviation is reported across three seeds where applicable.

Condition	Leaf	Top-5	Parent	H-Dist	Sev-W	ECE
Zero-shot CLIP	56.05	81.95	70.76	1.664	63.41	0.129
Flat adapter	73.15	93.46	83.12	0.972	78.14	0.025
Hierarchy, clean	73.20	93.39	79.34	0.960	78.33	0.018
Hierarchy, uniform 20% noise	72.56	92.96	78.65	0.998	77.60	0.069

5.2 Leaf-Accuracy Degradation by Noise Type

The central result is shown in Fig. 1 and Table 2. Uniform noise is surprisingly robust. Even when every training parent label is replaced by a random wrong parent, leaf accuracy remains 71.19%, only about two percentage points below the clean hierarchy-aware result. In-domain noise is more damaging, reducing leaf accuracy to 63.97% at full corruption. Cyclic noise is most harmful, dropping to 47.93% at full corruption.

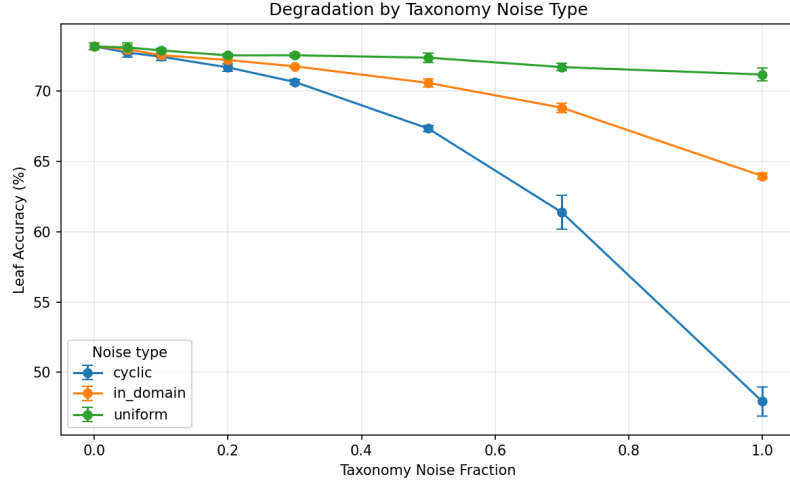


Fig. 1. Leaf-accuracy degradation by taxonomy noise type. Error bars show standard deviation across three seeds.

Table 2. Leaf accuracy under parent-label corruption.

Noise type	0%	20%	50%	100%
Uniform	73.20	72.56	72.39	71.19
In-domain	73.20	72.23	70.60	63.97
Cyclic	73.20	71.70	67.35	47.93

Figure 2 shows the same result as a delta from the clean hierarchy baseline. This representation makes the severity of systematic noise easier to read: cyclic corruption costs more than 25 percentage points at full corruption, while uniform corruption costs only about two percentage points. The delta plot is especially useful because the three curves share the same clean baseline at zero noise.

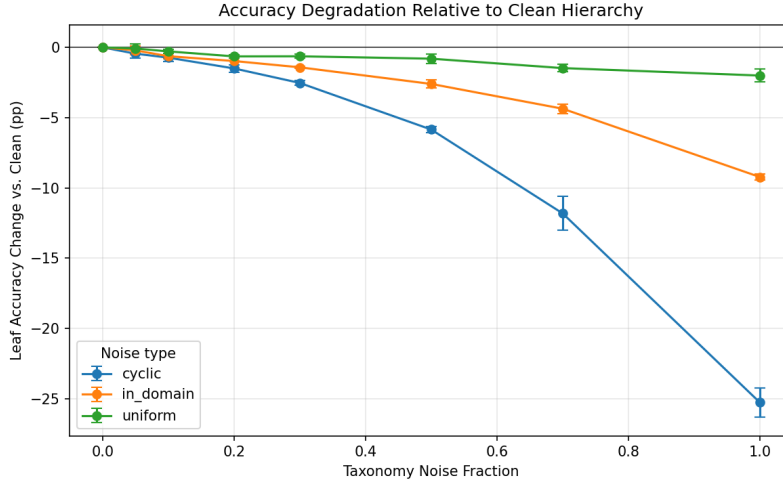


Fig. 2. Leaf-accuracy change relative to clean hierarchy-aware adaptation.

5.3 Calibration and Semantic Error Severity

Accuracy alone hides two important effects. First, noisy taxonomy supervision can harm calibration even when it does not strongly reduce top-1 accuracy. Figure 3 shows that uniform noise produces the steepest ECE increase, reaching 0.318 at full corruption. Thus, random parent corruption mostly preserves the ranking of the correct class, but it makes the model’s confidence less reliable.

Second, structured noise changes the semantic severity of errors. Figure 4 plots hierarchical distance against noise fraction. Cyclic noise causes the mean distance to grow sharply, indicating that errors move from near misses within related parents toward more severe cross-parent or cross-domain confusions. Uniform noise leaves hierarchical distance relatively stable, which is consistent with its smaller effect on leaf accuracy.

5.4 Parent Collapse Versus Leaf Robustness

Figure 5 compares parent accuracy and leaf accuracy across all noisy hierarchy-aware runs. The scatter shows that parent accuracy can collapse before leaf

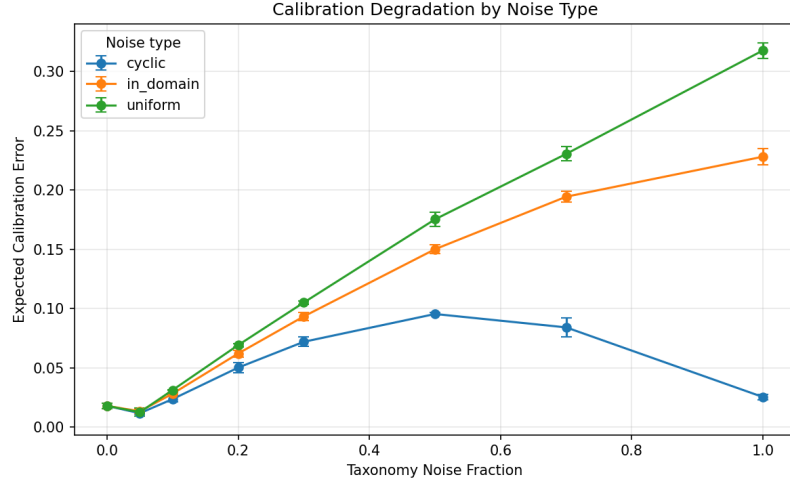


Fig. 3. Expected calibration error as parent-label corruption increases.

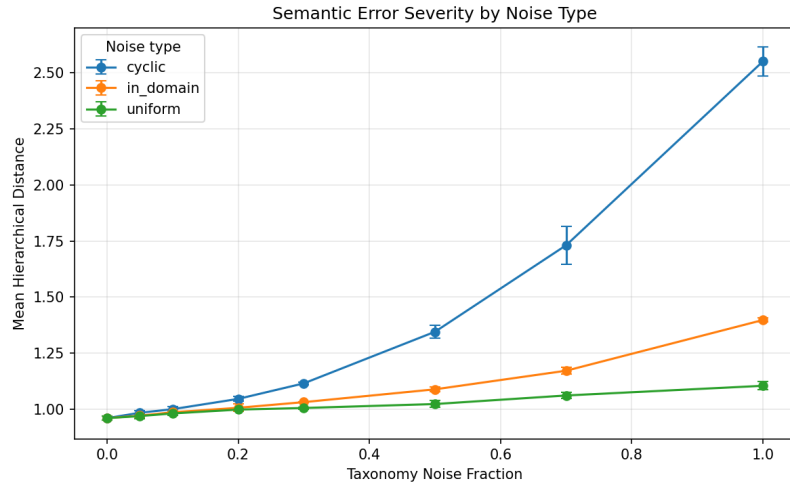


Fig. 4. Mean hierarchical distance, where larger values indicate semantically more severe errors.

accuracy fully collapses. Uniform full corruption yields near-zero parent accuracy while leaf accuracy remains near 71%. In contrast, cyclic corruption produces both low parent accuracy and a major leaf-accuracy drop. This supports the interpretation that parent-head failure is not by itself sufficient to destroy the leaf task; the damage is largest when the wrong parent signal is coherent enough to reshape the leaf boundary.

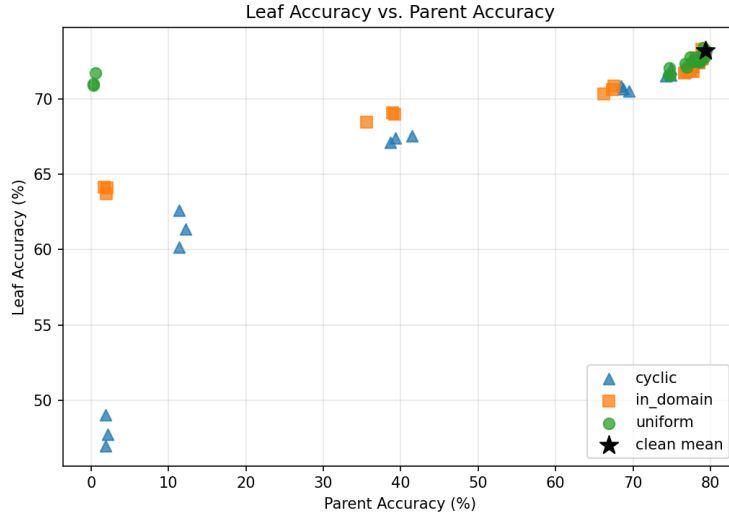


Fig. 5. Relationship between parent accuracy and leaf accuracy across noisy hierarchy-aware runs.

5.5 Per-Parent Sensitivity

Aggregate degradation curves do not show which parts of the taxonomy are most fragile. Figure 6 presents per-parent leaf accuracy under cyclic noise, the most damaging corruption type. The heatmap shows that accuracy degradation is not uniform. Parent groups such as aquatic mammals and small mammals become much darker at high corruption, while large man-made outdoor things and vehicles remain comparatively strong. This suggests that taxonomy noise interacts with visual separability: visually distinctive parent groups can retain useful leaf boundaries even when parent supervision is incorrect, whereas visually confusable animal groups are more vulnerable.

5.6 Interpretation

The results support two main interpretations. First, hierarchy-aware adaptation is not automatically superior to flat adaptation when leaf labels are abundant.

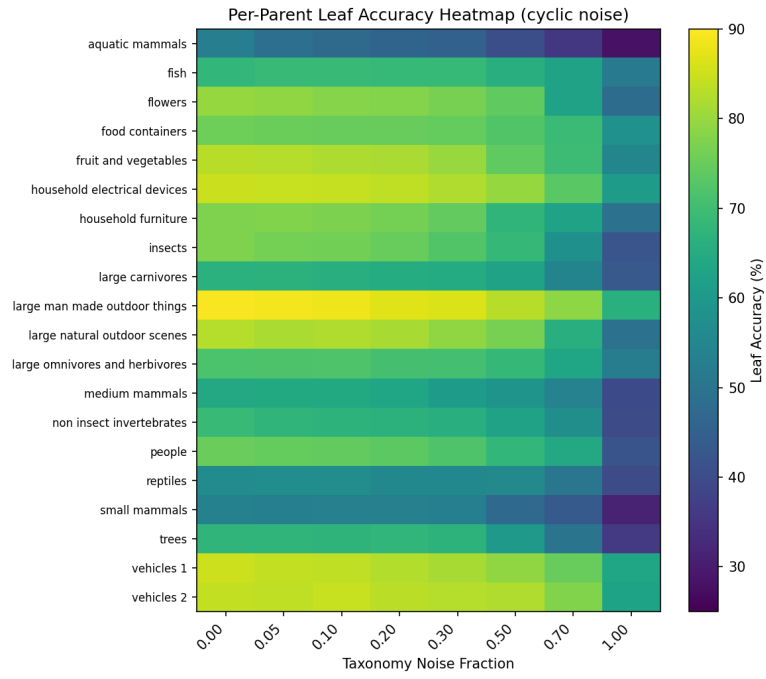


Fig. 6. Per-parent leaf accuracy under cyclic taxonomy noise. Each cell averages over three seeds.

On CIFAR-100, clean hierarchy supervision mainly improves secondary metrics such as hierarchical distance and calibration. Second, taxonomy-noise robustness depends heavily on whether errors are independent or structured. Uniform corruption distributes wrong parent labels across the taxonomy, so the auxiliary signal is noisy but not consistently misleading in one direction. In-domain corruption is semantically plausible and therefore more likely to interfere with boundaries among related visual groups. Cyclic corruption is coherent and systematic; the model receives a stable but wrong mapping between leaves and parents, which strongly damages the coupled leaf-to-parent consistency objective.

6 Reproducibility

The full source code, experiment scripts, generated results, figures, and Springer-format report are maintained in the GitHub repository:

<https://github.com/manuvikash/VLM-finetune>

The committed experiment artifacts include the aggregate result JSON, the per-parent metric CSV, the noise-type summary CSV, and all figures used in this paper. The generated JPEG image directory is intentionally ignored by git because it contains 60,000 reproducible CIFAR-derived files. It can be regenerated from the committed manifests and builder script.

The main commands are:

```
python build_dataset.py
  --num-coarse 20 --per-class 0
  --use-test-split

python feature_sweep.py
  --epochs 30 --seeds 42 123 456
  --noise-fractions 0.0 0.05 0.1 0.2
  0.3 0.5 0.7 1.0
  --noise-types uniform in_domain cyclic
  --lambda-parent 1.0
  --consistency-weight 0.8 --lr 0.001

python analysis.py
  --export-csv results
  --plot --plot-dir figures
  --per-parent-csv runs/per_parent_metrics.csv
```

7 Conclusion

This paper evaluated hierarchy-aware CLIP adaptation under controlled taxonomy noise. On the full CIFAR-100 hierarchy, clean hierarchy-aware adaptation

matches flat adaptation in leaf accuracy and modestly improves hierarchical distance and calibration. The main finding is that robustness depends strongly on the type of taxonomy error. Uniform parent-label noise is largely tolerated, but structured noise can severely harm fine-grained recognition. At full corruption, uniform noise retains 71.19% leaf accuracy, in-domain noise drops to 63.97%, and cyclic systematic noise drops to 47.93%. Additional visualizations show that calibration can degrade even when accuracy is stable, hierarchical distance exposes semantically worse mistakes under systematic noise, parent accuracy can collapse before leaf accuracy does, and different parent groups have different sensitivity to taxonomy corruption.

For practical VLM fine-tuning, the most important question is therefore not only how many parent labels are noisy, but whether the errors encode a coherent wrong taxonomy. Future work should repeat the study with full end-to-end backbone fine-tuning, larger VLMs, deeper real-world taxonomies, and asymmetric noise processes based on observed annotation errors. Another useful direction is to design hierarchy-aware losses that can detect and down-weight suspicious parent targets while preserving the benefits of clean hierarchical supervision.

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